512 steps: paired metric change vs anchor
Same LR, split, train/evaluation seed and weights; negative = improvement

	Nearest	Lateral	Angle	Length	Width	Speed	Collision
heading_l1 | S=0.906 | Nseed=1	-2.5%	+3.6%	+4.0%	+151.9%	-55.1%	-35.2%	-4.5%
balanced32_vector | S=0.921 | Nseed=1	-17.9%	+19.8%	-20.8%	+250.4%	+20.1%	+0.2%	+2.6%
anchor | S=0.928 | Nseed=1	+0.0%	+0.0%	+0.0%	+0.0%	+0.0%	+0.0%	+0.0%
legacy | S=0.942 | Nseed=1	-23.9%	+5.1%	+10.5%	+885.2%	+463.7%	-25.3%	-34.3%
geometry_strong | S=0.949 | Nseed=1	-7.5%	+3.6%	+6.8%	+223.4%	-65.8%	+82.9%	-7.8%
collision_weak | S=0.963 | Nseed=1	-5.3%	+3.4%	+10.0%	+142.7%	-2.3%	-20.3%	+2.5%
heading_p2 | S=0.967 | Nseed=1	-6.2%	+1.7%	+38.9%	+31.7%	-67.5%	-31.5%	-5.7%
warm_winner | S=0.967 | Nseed=1	-2.9%	+6.2%	+34.7%	+68.7%	+87.8%	+5.8%	-16.8%
position_mse | S=0.970 | Nseed=1	-12.3%	+6.3%	+9.2%	+70.0%	-21.0%	-20.0%	+7.3%
balanced32_p2 | S=0.982 | Nseed=1	+0.0%	+4.8%	-7.4%	+430.3%	+514.6%	-22.2%	-14.0%
speed_p2 | S=0.996 | Nseed=1	-0.4%	+5.4%	+5.0%	+367.4%	+258.5%	-8.3%	-6.2%
balanced32 | S=1.027 | Nseed=1	-14.1%	+20.3%	-28.3%	+265.9%	+7.3%	+83.8%	+22.5%
balanced32_speed_l1 | S=1.049 | Nseed=1	-7.7%	+27.3%	-25.1%	+88.2%	+76.6%	-17.0%	+33.1%
geo_l1_16 | S=1.061 | Nseed=1	-6.6%	+20.8%	+1.0%	+403.1%	-35.7%	-24.0%	+23.1%
pos_mse32 | S=1.080 | Nseed=1	-16.0%	+22.0%	-11.0%	+203.0%	+91.4%	+35.9%	+27.5%
width_p2 | S=1.104 | Nseed=1	-4.2%	+2.2%	+13.5%	+1140.1%	+555.0%	+25.5%	-15.3%
vector_heading | S=1.118 | Nseed=1	-6.5%	+2.7%	+18.4%	+770.2%	+633.9%	-19.1%	-6.7%
balanced32_shape_l1 | S=1.128 | Nseed=1	-16.3%	+29.8%	-20.7%	+411.8%	+74.5%	-0.4%	+42.8%
heading_p3 | S=1.134 | Nseed=1	-5.2%	+0.7%	+57.9%	+243.5%	+334.5%	+2.4%	-2.9%
geo_mse32 | S=1.141 | Nseed=1	-14.8%	+26.2%	-27.0%	+1113.5%	+195.8%	+104.2%	+20.5%
geo_mse64 | S=1.156 | Nseed=1	-7.0%	+18.8%	+2.2%	+585.2%	+252.7%	+23.7%	+23.7%
balanced32_p0 | S=1.292 | Nseed=1	-1.1%	+35.4%	-7.5%	+413.8%	+219.9%	+72.8%	+55.3%

Percent change; colors clipped ±100%
100
75
50
25
0
−25
−50
−75
−100